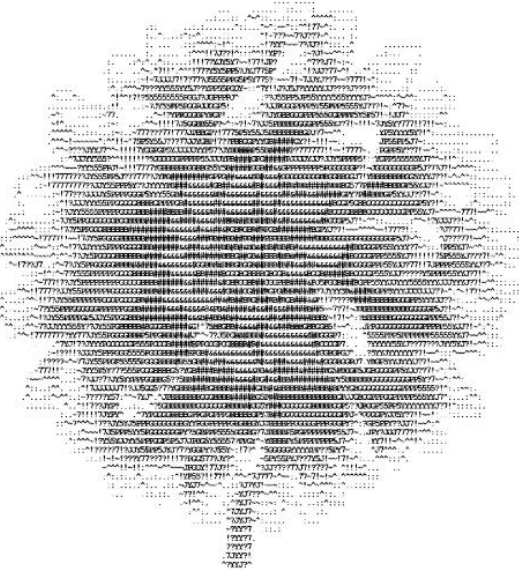


The False Promise of ChatGPT: Uncovering the Truth Behind Large Language Models

The New York Times

Noam Chomsky: The False Promise of ChatGPT

March 8, 2023



The New York Times

You Can Have the Blue Pill or the Red Pill, and We're Out of Blue Pills

March 24, 2023

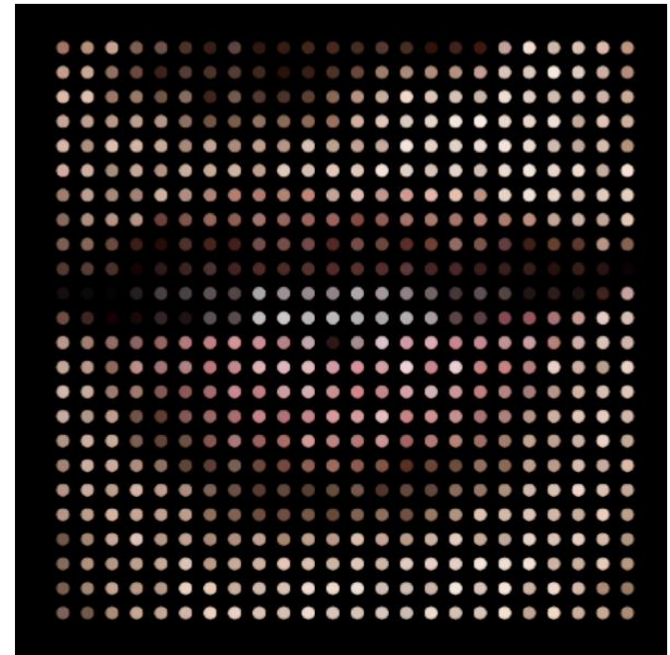
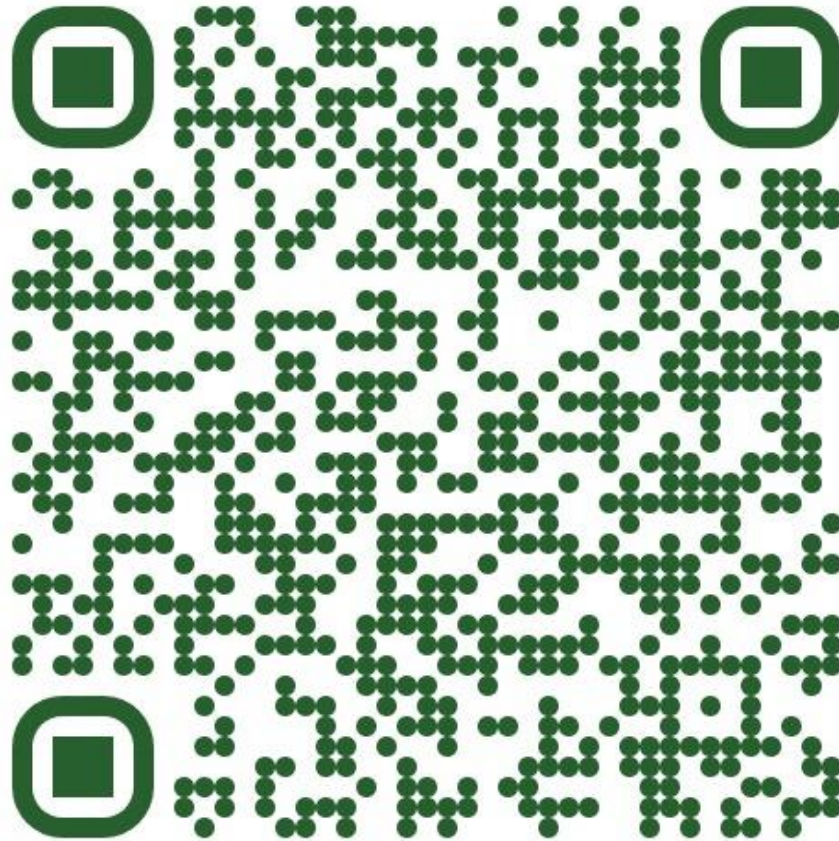


Illustration by Zisiga Mukulu/The New York Times; photograph by Jonathan Knowles/Getty Images

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What is a Large Language Model (LLM)?

- ❑ A Large Language Model (LLM), is a type of neural network.
- ❑ Designed to understand, generate, and respond to text in a human-like manner.
- ❑ Trained on massive amounts of text data, sometimes including large portions of the publicly available text on the internet.
- ❑ The term "large" refers to the model's size in terms of:
 - the number of parameters
 - the volume of the dataset
- ❑ LLMs use transformer architecture, which enables selective attention to various input texts, allowing them to effectively handle the complexities of human language



The LLM is simply (pre)trained model to predict the next word



What is a Large Language Model (LLM)?

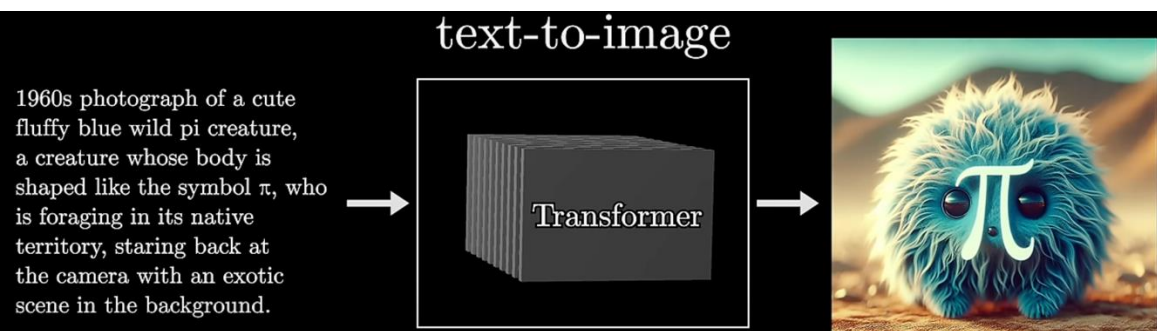
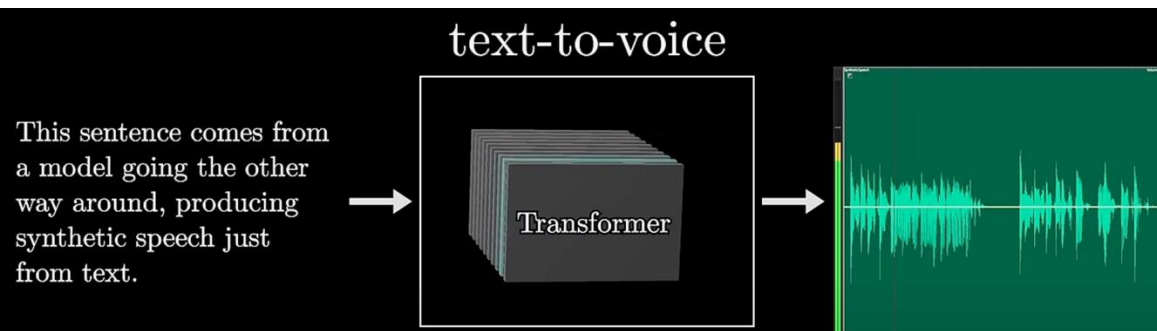
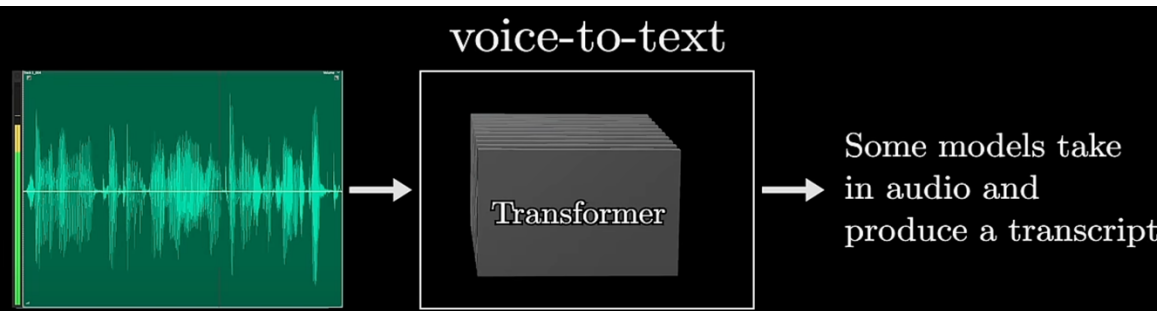


Image Source: <https://www.3blue1brown.com/>

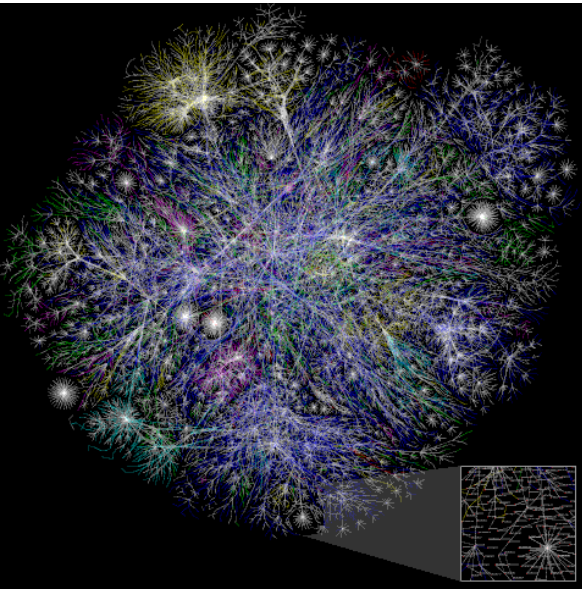


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What is a Large Language Model (LLM)?

Llama 2 70B Model

Chunk of the internet



6,000 GPUs for 12 days, ~\$2M



~140GB file
is the compression of the
internet

parameters

Image Source:
<https://www.cryptopolitan.com/openai-chief-pursues-trillions-in-investment/wikipedia.org>

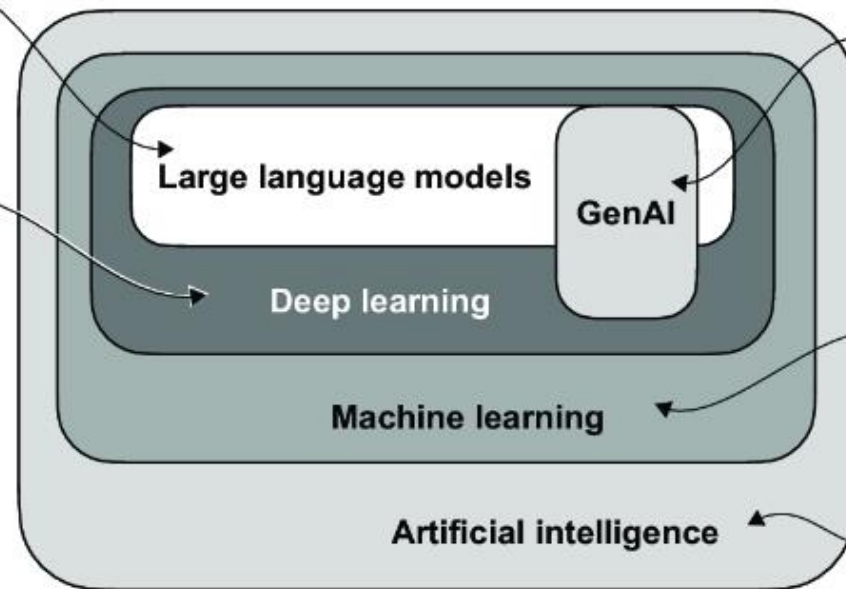


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What is a Large Language Model (LLM)?

Deep neural network for parsing and generating human-like text

Machine learning with neural networks consisting of many layers

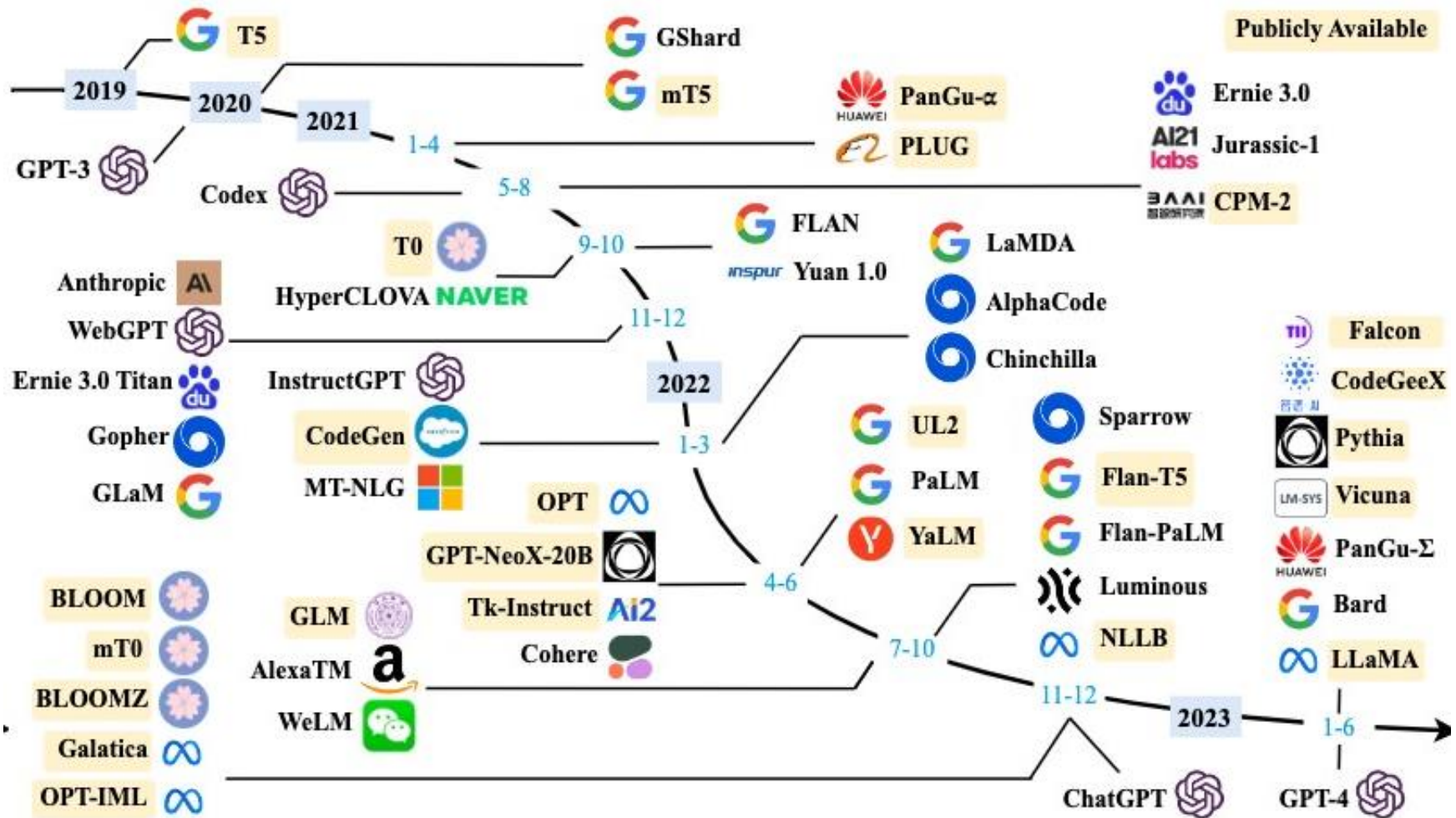


GenAI involves the use of deep neural networks to create new content, such as text, images, or various forms of media

Algorithms that learn rules automatically from data

Systems with human-like intelligence

What is a Large Language Model (LLM)?



Neural Network

What a computer sees!

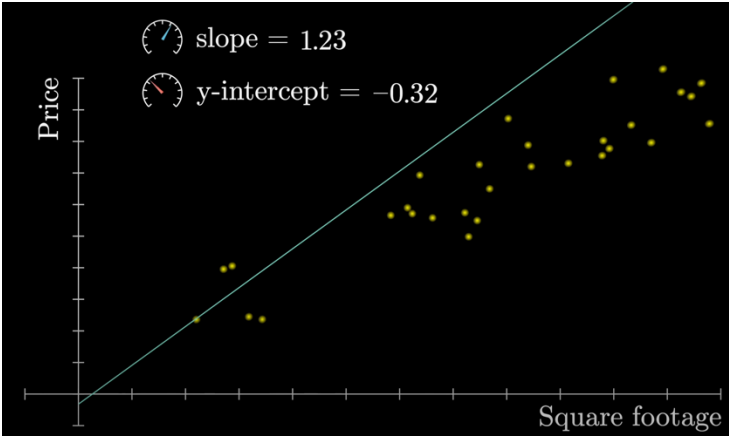
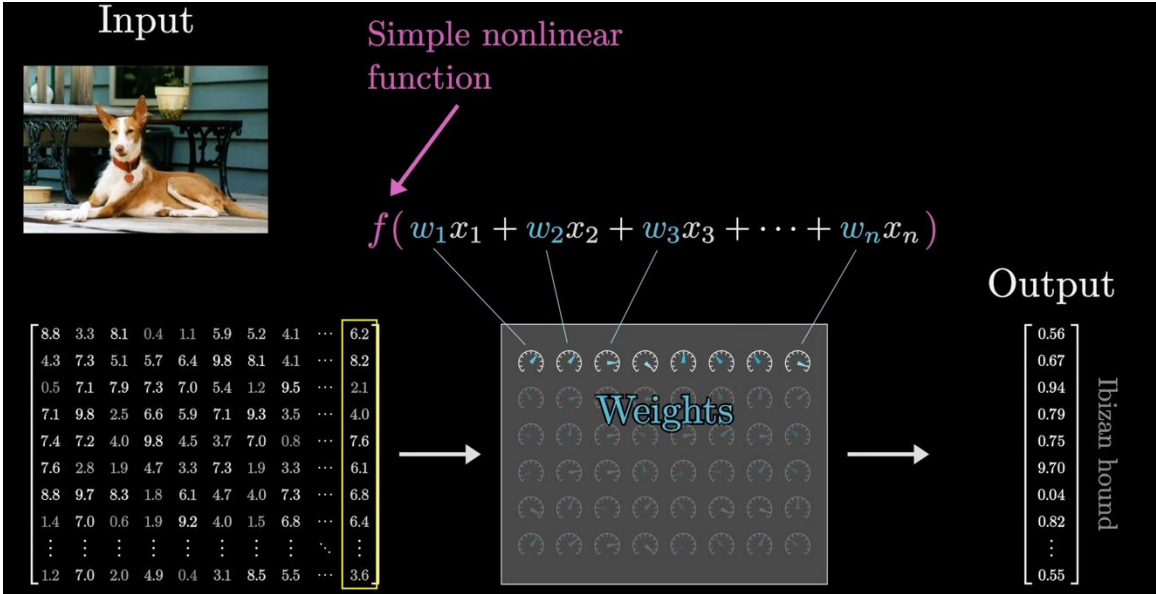
Annotated Pixel Values

160	155	143	154	176	158	123	114	131	136
147	161	161	155	118	112	141	161	155	148
151	148	127	99	149		143	134	135	147
148	124	143	177	147	117	146	148	147	128
103	163		107	150	164	140	115	112	104
151	106	79	124	131	121	91	93	112	147
74	86	118	114	90	68	106	111	125	72
111	103	78	93	66	107	103	73	80	83
83	89	104	64	99	72	46	76	76	72
71	90	69	71	63	56	82	64	68	90

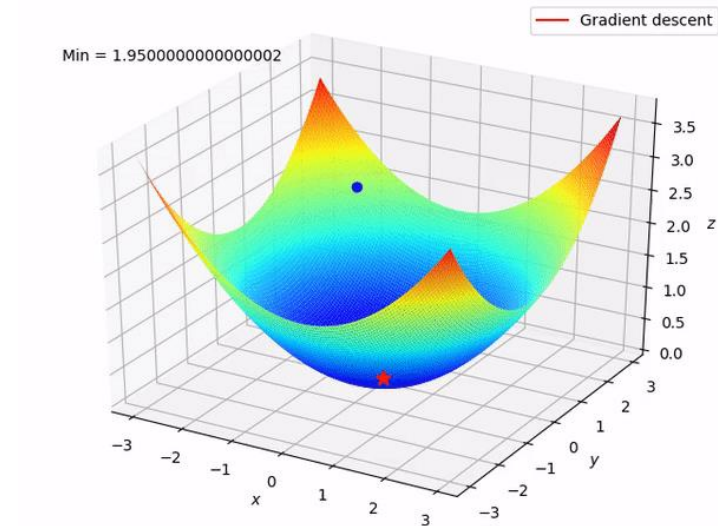
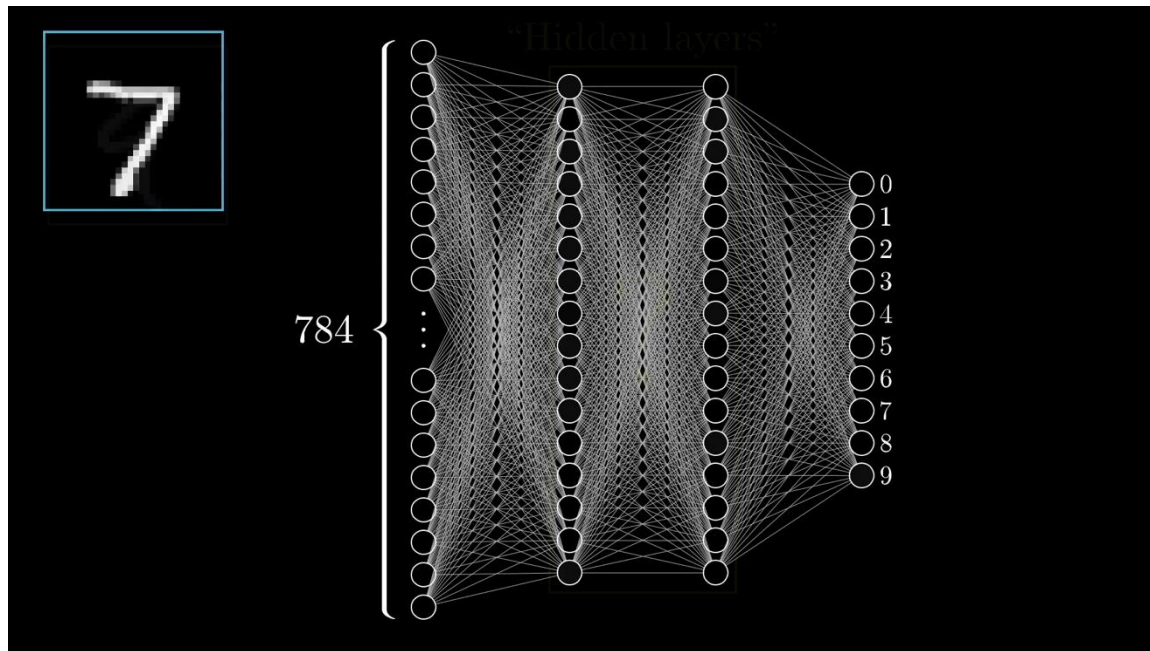
What a human sees!



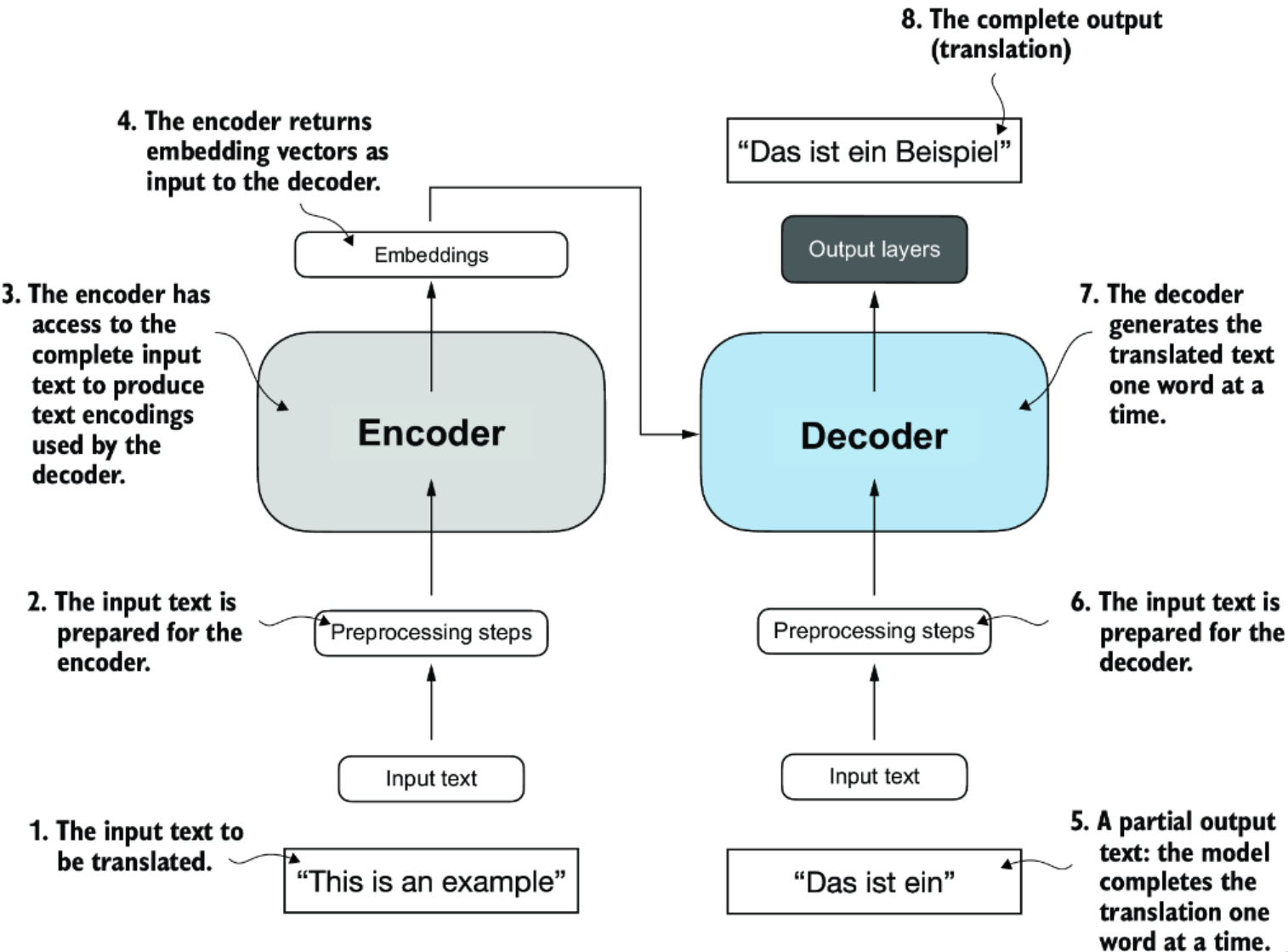
Neural Network



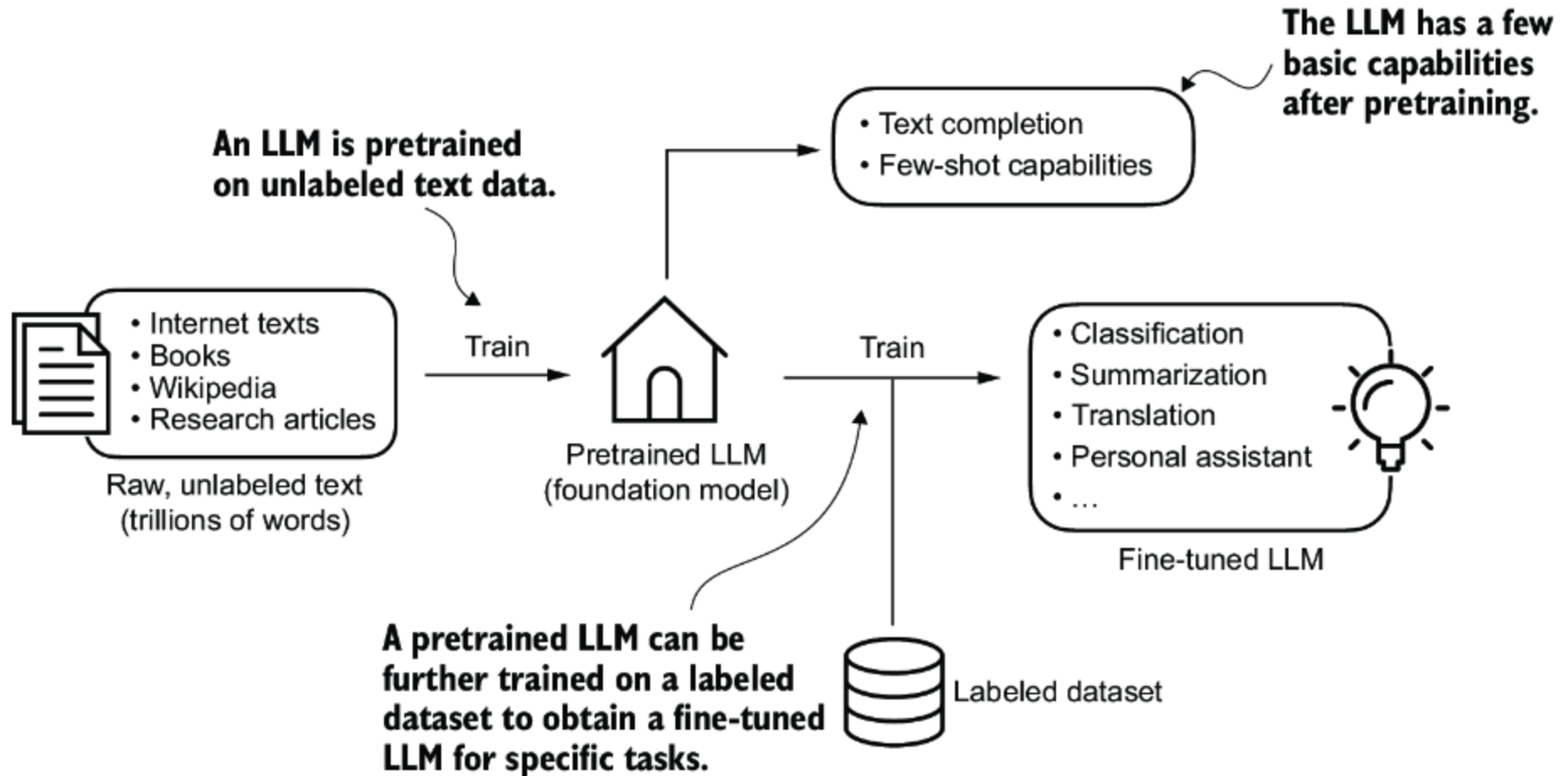
Neural Network



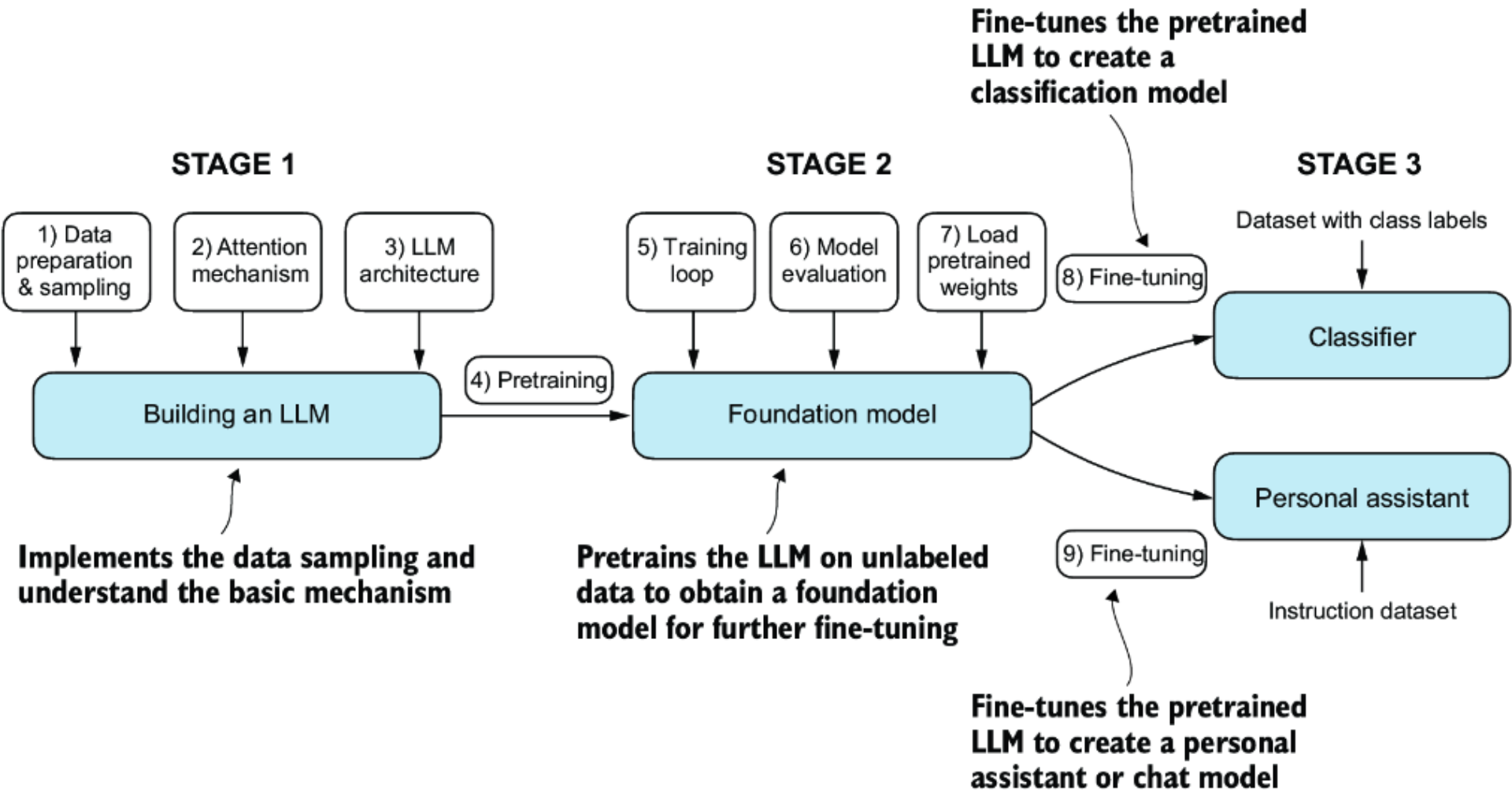
Transformer Architecture



Stages of Building and Using LLMs



Stages of Building and Using LLMs



Stage 1: Building

Table: The pretraining dataset of the popular GPT-3 LLM

Dataset	Dataset description	Number of tokens	Proportion in training data
CommonCrawl (filtered)	Web crawl data	410 billion	60%
WebText2	Web crawl data	19 billion	22%
Books1	Internet-based book corpus	12 billion	8%
Books2	Internet-based book corpus	55 billion	8%
Wikipedia	High-quality text	3 billion	3%



Stage 1: Building

Preparing Text for LLM Training

Next word prediction forces the neural network to learn extensively about the world

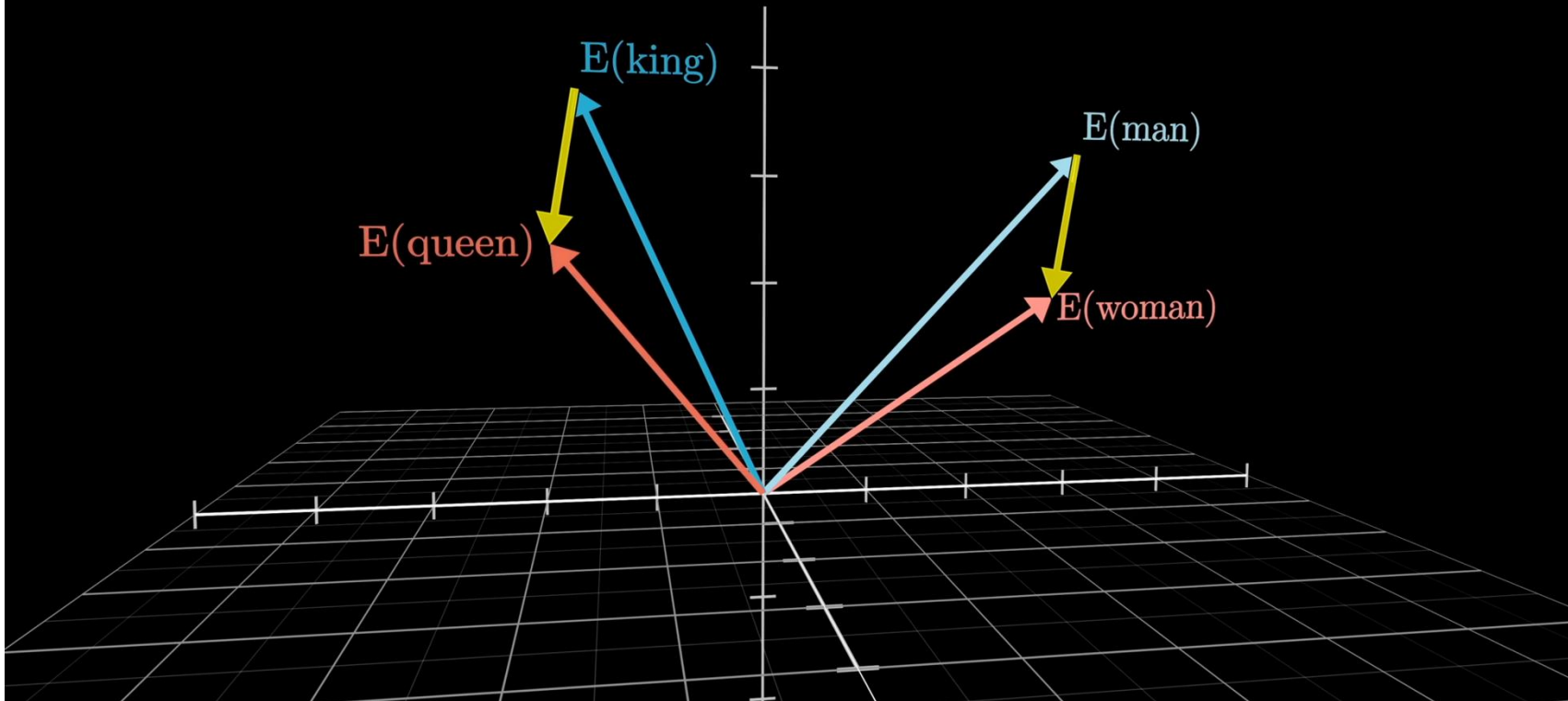
	Input Tokens	Target Token_IDS	Input Context	Target Token
0	[12211]	3854	Second	Law
1	[12211, 3854]	286	Second Law	of
2	[12211, 3854, 286]	47061	Second Law of	Robotics
3	[12211, 3854, 286, 47061]	25	Second Law of Robotics	:
4	[12211, 3854, 286, 47061, 25]	317	Second Law of Robotics :	A
5	[12211, 3854, 286, 47061, 25, 317]	9379	Second Law of Robotics : A	robot
6	[12211, 3854, 286, 47061, 25, 317, 9379]	1276	Second Law of Robotics : A robot	must
7	[12211, 3854, 286, 47061, 25, 317, 9379, 1276]	22389	Second Law of Robotics : A robot must	obey
8	[12211, 3854, 286, 47061, 25, 317, 9379, 1276, 22389]	262	Second Law of Robotics : A robot must obey	the
9	[12211, 3854, 286, 47061, 25, 317, 9379, 1276, 22389, 262]	6266	Second Law of Robotics : A robot must obey the	orders
10	[12211, 3854, 286, 47061, 25, 317, 9379, 1276, 22389, 262, 6266]	1813	Second Law of Robotics : A robot must obey the orders	given
11	[12211, 3854, 286, 47061, 25, 317, 9379, 1276, 22389, 262, 6266, 1813]	340	Second Law of Robotics : A robot must obey the orders given	it



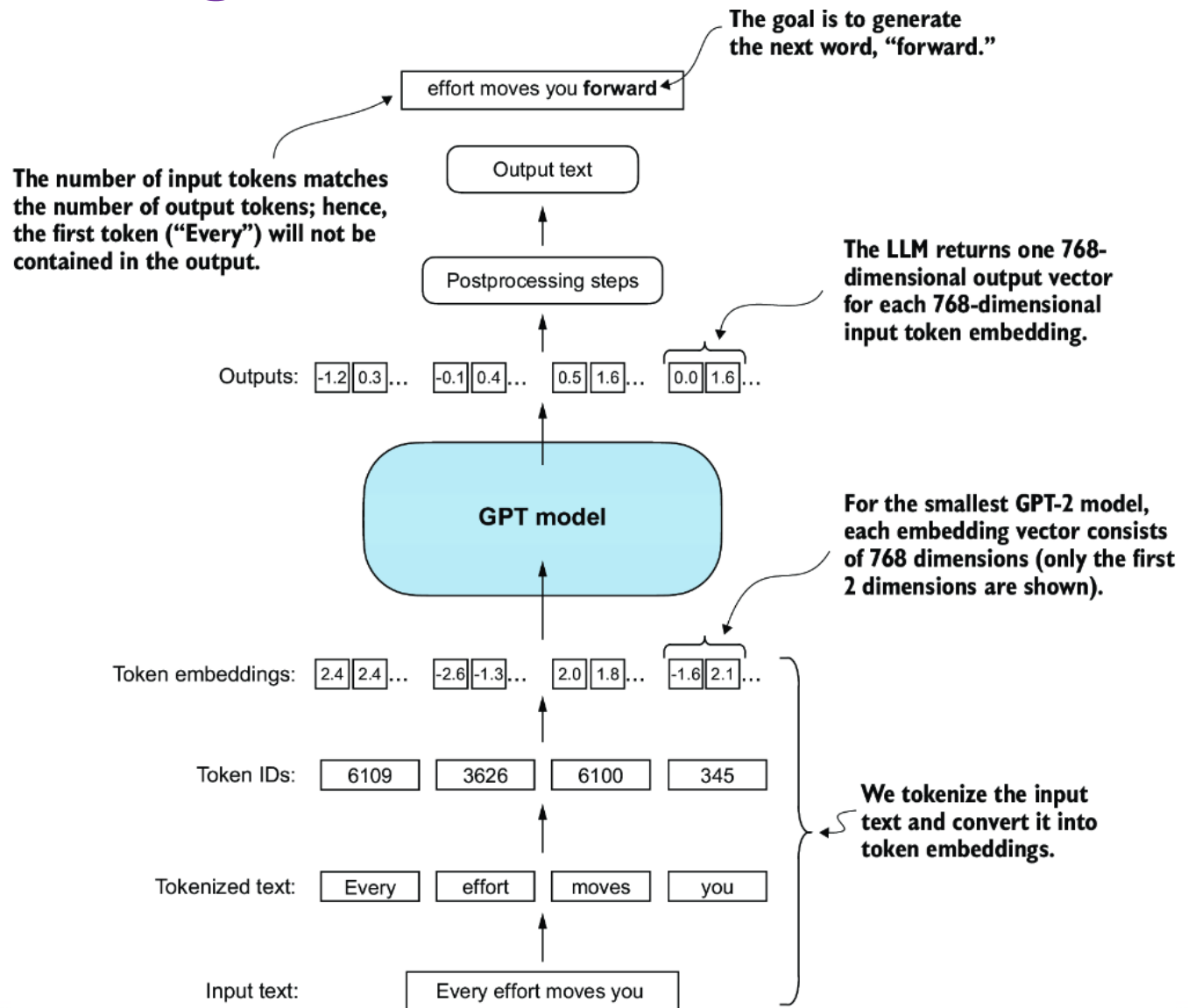
Stage 1: Building

Understanding word embeddings

$$E(\text{queen}) \approx E(\text{king}) + E(\text{woman}) - E(\text{man})$$



Stage 1: Building



Stage 1: Building

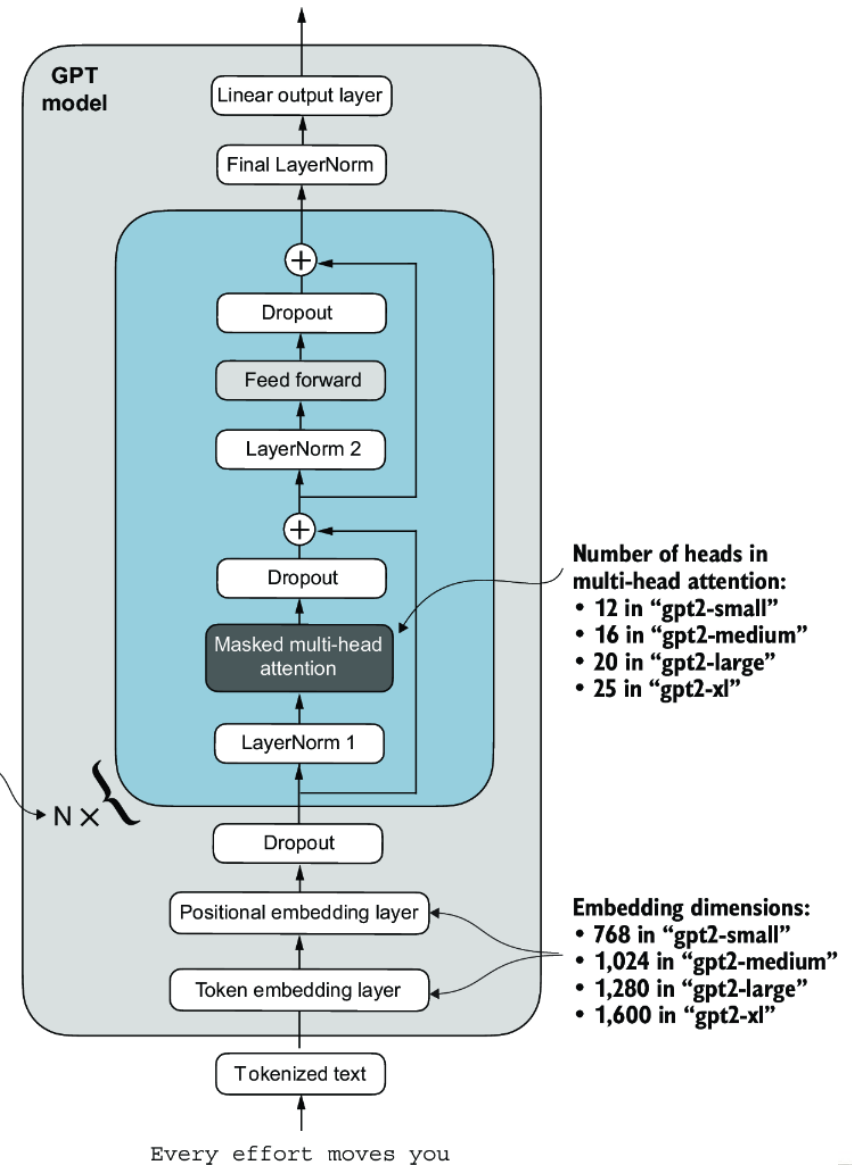
LLM architectures

Total number of parameters:

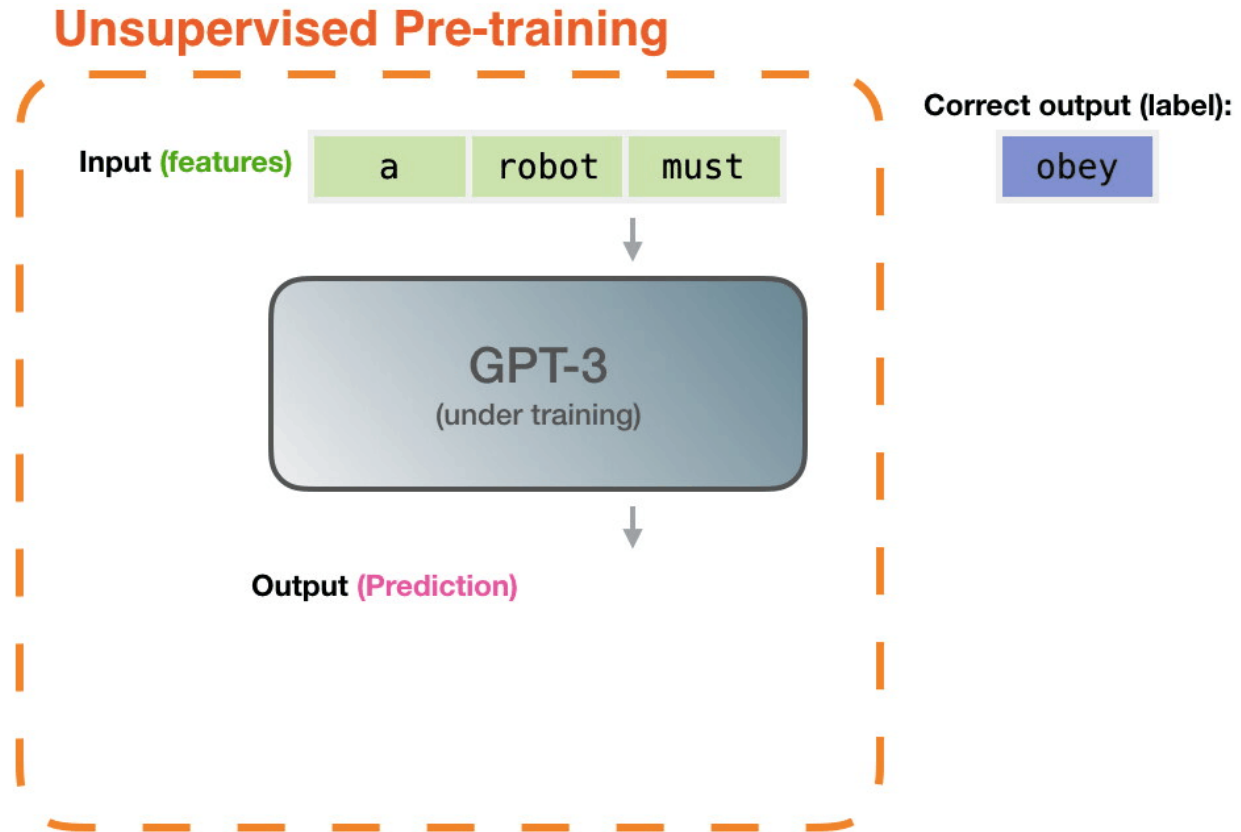
- 124 M in “gpt2-small”
- 355 M in “gpt2-medium”
- 774 M in “gpt2-large”
- 1558 M in “gpt2-xl”

Repeat this transformer block:

- 12 × in “gpt2-small”
- 24 × in “gpt2-medium”
- 36 × in “gpt2-large”
- 48 × in “gpt2-xl”

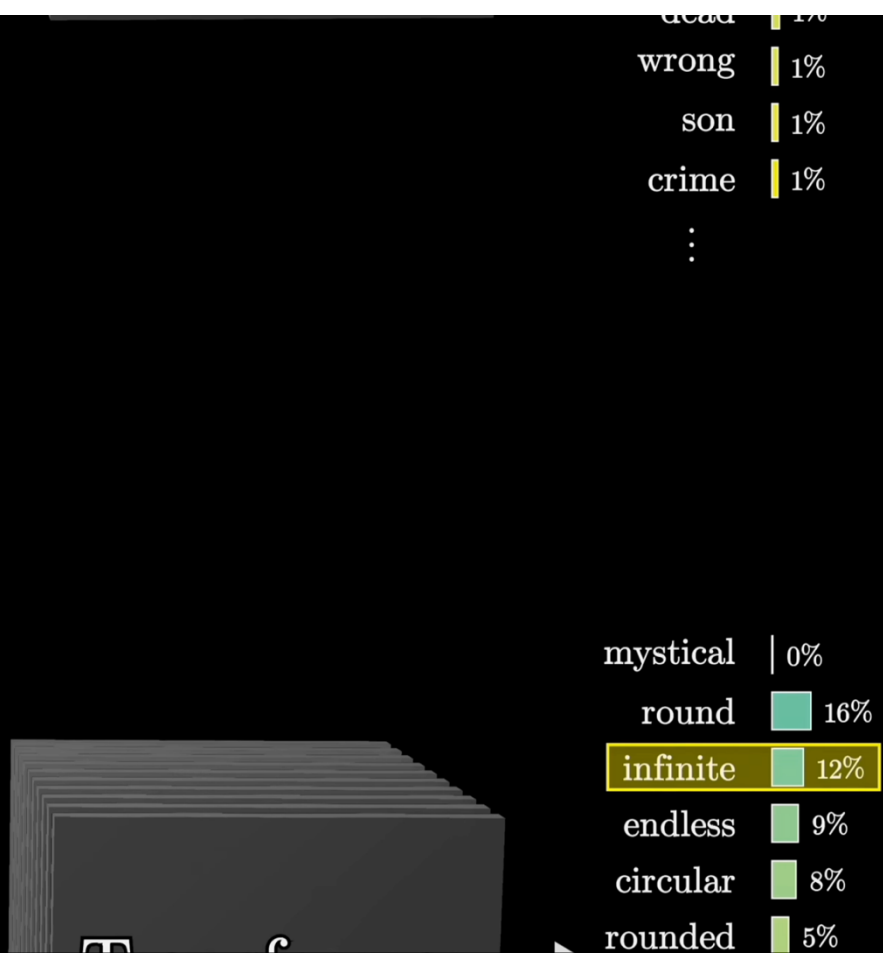


Stage 2: Pretraining



How do LLMs Generate Sequences of Words?

Behold, a wild pi creature,
foraging in its native habitat of
mathematical formulas and
computer code! With its infinite



Stage 2: Pretraining

Rank* (UB)	Model	Arena Score	95% CI	Votes	Organization	License	Knowledge Cutoff
1	o1-preview	1355	+12/-11	2991	OpenAI	Proprietary	2023/10
2	ChatGPT-4o-latest (2024-09-03)	1335	+5/-6	10213	OpenAI	Proprietary	2023/10
2	o1-mini	1324	+12/-9	3009	OpenAI	Proprietary	2023/10
4	Gemini-1.5-Pro-Exp-0827	1299	+5/-4	28229	Google	Proprietary	2023/11
4	Grok-2-08-13	1294	+4/-4	23999	xAI	Proprietary	2024/3
6	GPT-4o-2024-05-13	1285	+3/-3	90695	OpenAI	Proprietary	2023/10
7	GPT-4o-mini-2024-07-18	1273	+3/-3	30434	OpenAI	Proprietary	2023/10
7	Claude 3.5 Sonnet	1269	+3/-3	62977	Anthropic	Proprietary	2024/4
7	Gemini-1.5-Flash-Exp-0827	1269	+4/-4	22264	Google	Proprietary	2023/11
7	Grok-2-Mini-08-13	1267	+4/-5	22041	xAI	Proprietary	2024/3
7	Gemini Advanced App (2024-05-14)	1267	+3/-3	52218	Google	Proprietary	Online
7	Meta-Llama-3.1-405b-Instruct-fp8	1266	+4/-4	31280	Meta	Llama 3.1 Community	2023/12
7	Meta-Llama-3.1-405b-Instruct-bf16	1264	+6/-8	5865	Meta	Llama 3.1 Community	2023/12
8	GPT-4o-2024-08-06	1263	+4/-3	22562	OpenAI	Proprietary	2023/10
12	Gemini-1.5-Pro-001	1259	+3/-3	80656	Google	Proprietary	2023/11
14	GPT-4-Turbo-2024-04-09	1257	+3/-2	92973	OpenAI	Proprietary	2023/12
17	GPT-4-1106-preview	1251	+2/-3	97378	OpenAI	Proprietary	2023/4
17	Mistral-Large-2407	1250	+4/-4	27285	Mistral	Mistral Research	2024/7
17	Athena-70b	1250	+4/-4	20738	NexusFlow	CC-BY-NC-4.0	2024/7

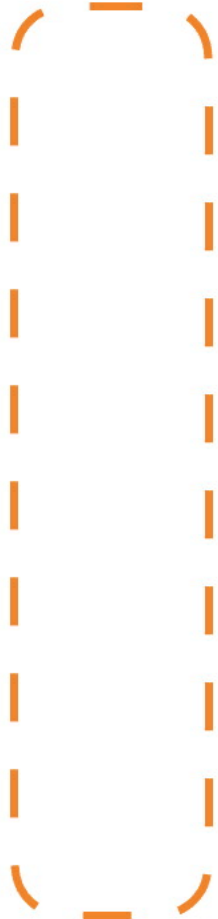
Image Source: <https://lmarena.ai/>



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Stage 3: Finetuning

Pre-training

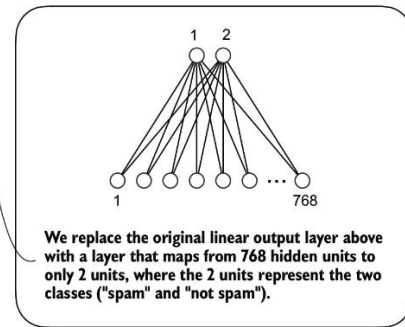
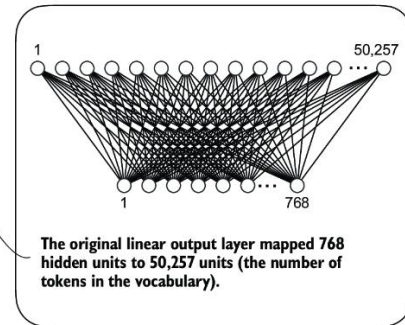
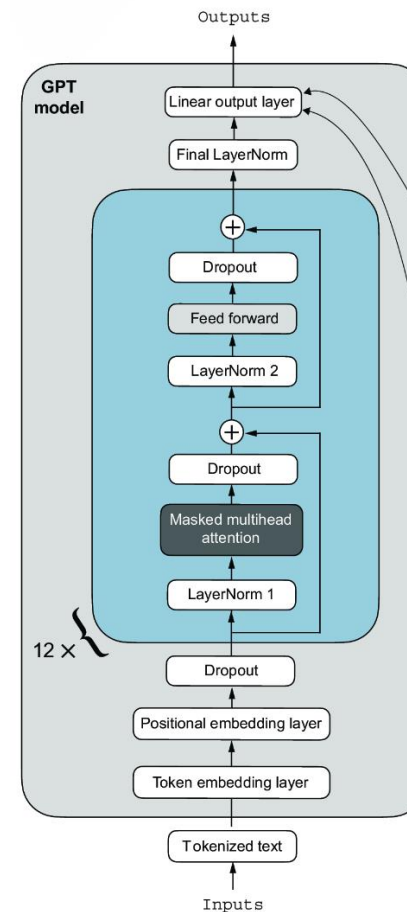
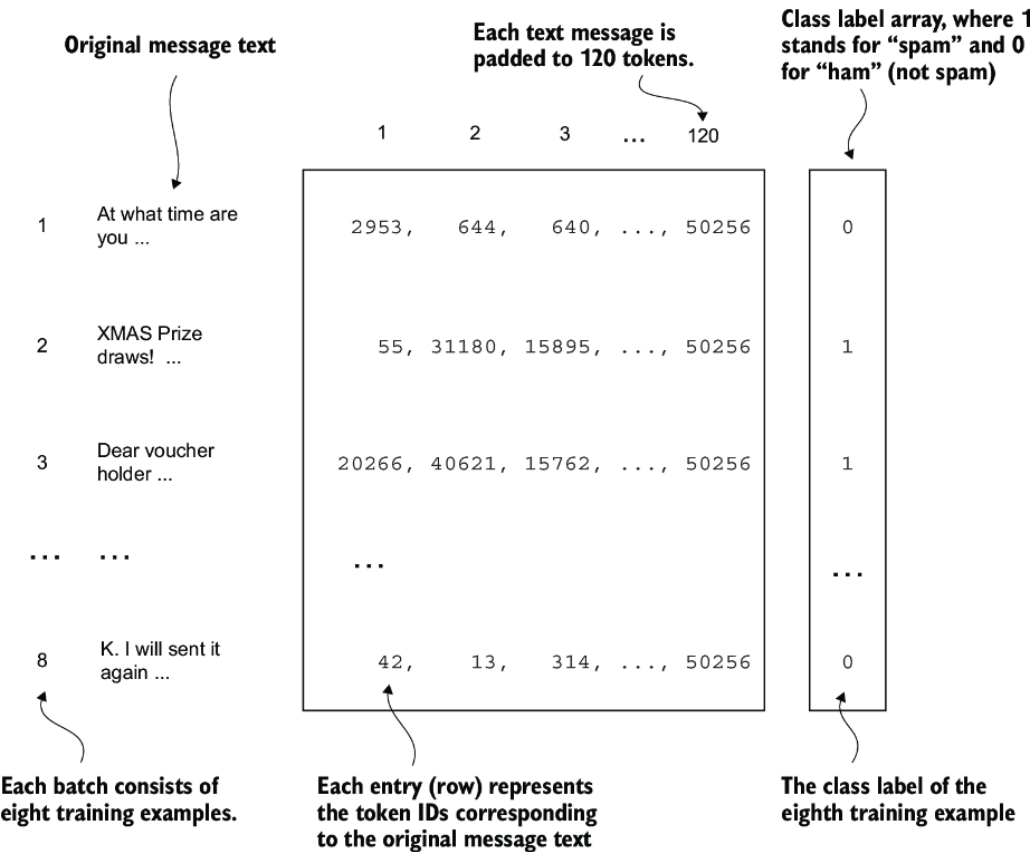


Fine-tuning

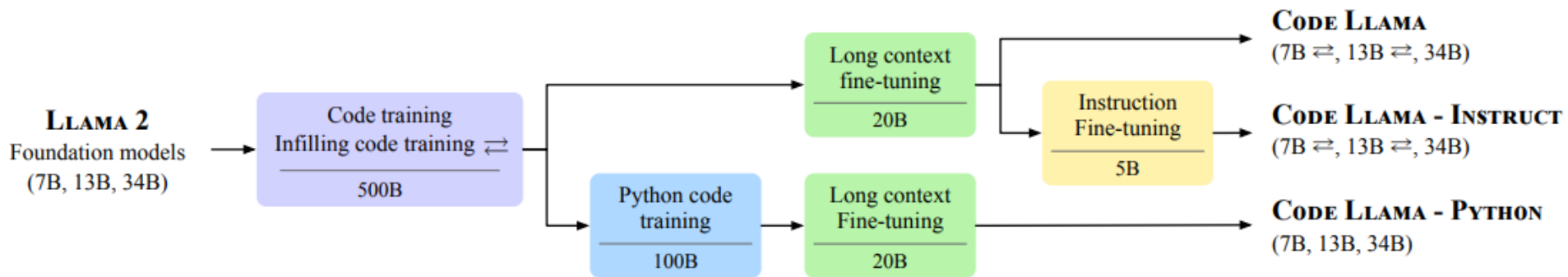
Additional training to become better at a certain task

Example: English to French Translation

Stage 3: Finetuning



Stage 3: Finetuning



Summary

Stage 1 and 2: Building LLM and Pretraining

1. Download ~10TB of text.
2. Get a cluster of ~6,000 GPUs.
3. Compress the text into a neural network, pay ~\$2M, wait ~12 days.
4. Obtain base model

Stage 3: Finetuning

1. Write labeling instructions
2. Hire people, collect 100K high quality ideal Q&A responses.
3. Finetune base model on this data, wait ~1 day.
4. Obtain assistant model.
5. Run a lot of evaluations.
6. Deploy
7. Monitor, collect misbehaviors, go to step 1



The Transformer!!

Attention Is All You Need

by Google Brain (2017)

